

Evaluating UWB TWR-Based Localization for Lunar Rovers Under Physical Obstructions

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Abstract—Autonomous lunar construction rovers require centimeter-level localization to navigate the unstructured lunar terrain. Because satellite navigation is unavailable and dead reckoning is prone to rapid drift, Ultra-Wideband (UWB) two-way ranging has emerged as a promising Local Positioning System (LPS). However, UWB’s accuracy under physical line-of-sight obstructions such as regolith and metal barriers remains poorly quantified. This paper presents a controlled benchtop study evaluating a four-anchor, single-tag UWB network to determine how lunar-relevant obstructions degrade 3D localization. A stationary tag was tracked while a single anchor link was blocked by one of five materials spanning density and reflectivity: an empty container, the same container filled with sand (a regolith analog) or wrapped in aluminum foil, a paver stone, and a concrete block. Results demonstrate that single-link obstruction raised the 3D root-mean-square error (RMSE) from a 5.1 cm baseline to 13.0 cm for foil, 17.3 cm for sand, and 47.8 cm for concrete. This degradation is driven almost entirely by excess propagation delay; the obstruction injects a positive range bias on the blocked link that translates into position error at a near-unity slope of 0.998 ($r = 0.92$). Signals were never lost; the failure mode is the solver’s silent incorporation of the delayed range. These findings emphasize the necessity of integrating obstruction detection and geometrically balanced anchor placements for reliable autonomous lunar operations.

Index Terms—ultra-wideband, two-way ranging, indoor localization, signal obstruction, non-line-of-sight, lunar navigation, trilateration

I. INTRODUCTION

Traditional, large autonomous robots on Earth rely on established infrastructure, such as Global Positioning Systems (GPS) and cellular networks, to navigate complex environments and stream data back to a central location. However, as space exploration transitions from short-term orbital missions to surface exploration and permanent bases, fully autonomous construction rovers deployed on the lunar surface will face a highly unstructured environment lacking any of the aforementioned infrastructure. Currently, lunar surface vehicles employ dead reckoning (tracking how far they have moved since their initial placement) paired with inertial measurement units (IMUs) and visual odometry [1]. Although functional for exploration and short-term missions, these technologies eventually accumulate localization errors over time and distance, especially on the lunar surface, where light changes and the rover’s wheels can slip on the regolith. Constructing a permanent lunar base accurately and autonomously requires a

reliable, easily deployable, centimeter-level localization system that does not rely on satellites. Currently, no satellite navigation constellation serves the lunar surface. Although lunar communication/navigation infrastructure has been proposed [2], such satellite-based systems remain costly to deploy and would not provide the accuracy required to build a lunar base properly. Ultra-Wideband (UWB) radio is a promising solution to this need. Unlike traditional protocols like Wi-Fi and Bluetooth, which calculate approximate distances and positions based on rough signal strength, UWB systems utilize high-bandwidth, short pulses of data to enable Time-of-Flight (ToF) ranging. ToF ranging measures the amount of time a signal takes to go round trip between a tag and an anchor to determine the distance they have from each other. UWB localization systems have already been proven to work indoors on Earth with consistent, centimeter-level accuracy [3]. By implementing Two-Way Ranging (TWR) protocols [4], a moving tag (such as one mounted on an autonomous rover) actively requests a response from a grid of stationary anchors to determine the exact transmission time, and therefore its distance from each anchor. Besides UWB’s high accuracy, it has also proven to be very resistant to multipath interference [3]. Multipath interference occurs when radio signals bounce off terrain and other structures before reaching their destination. Because UWB transmits extremely short pulses of data, receivers can distinguish between direct responses and delayed or echoed responses. This resistance establishes it as a strong candidate for creating a Local Positioning System (LPS) on the Moon for lunar construction and other autonomous rovers. Despite extensive testing in terrestrial environments and UWB’s proven upsides, not enough testing has been done to emulate physical line-of-sight obstructions on the Moon. The lunar surface is filled with highly irregular terrain, intense environmental conditions, and other factors making it very different from our environment on the Earth. Prior work has proposed filtering-based mitigations for non-line-of-sight conditions [5], but without quantifying the underlying degradation by obstruction material. However, to successfully implement a reliable UWB network on the Moon, the effects of different kinds of obstructions on a UWB system must be quantified. Different kinds of obstructions like regolith and reflective materials on the Moon will impact the signal strength (attenuation) or the speed at which it travels (propagation

delay) in different ways [6]. An understanding of each material and its effect will help us design systems and algorithms to avoid these problems and compensate for any residual issues. Therefore, the purpose of this research is to investigate how physical obstructions affect the accuracy of UWB TWR-Based localization for construction vehicles or rovers on the Moon. By measuring the discrepancy between the tag's true position and the coordinates calculated by the trilateration algorithm under various obstructions, the study should be able to quantify the vulnerability of UWB networks under physical barriers. Two predictions follow. First, at fixed in-path depth, ranging error should increase with the bulk density of the obstruction. Second, high surface reflectivity should independently increase ranging error, irrespective of density.

II. LOCALIZATION MODEL

To account for any rough terrain or elevation changes, the system localizes the rover in three dimensions. Each anchor i at the known position $\mathbf{a}_i = (x_i, y_i, z_i)$ constrains the unknown tag position $\mathbf{p} = (x, y, z)$ to somewhere on the surface of a sphere with its radius equal to the measured range d_i . This is given by the geometric equation for a sphere:

$$(x - x_i)^2 + (y - y_i)^2 + (z - z_i)^2 = d_i^2. \quad (1)$$

Closed-form solutions to this system exist, but they handle over-determined, noisy range sets poorly and do not admit per-link weighting. The tag therefore utilizes a best-fit algorithm in its solver. It searches for the coordinate $\mathbf{p} = (x, y, z)$ that best aligns with all of the measured ranges at once (as close as possible to being on every sphere). For any candidate position, the error to anchor i is the residual $r_i(\mathbf{p}) = \|\mathbf{p} - \mathbf{a}_i\| - d_i$, the gap between the distance that candidate implies and the distance actually measured. The tag will then use the position value that makes the total squared error across all n anchors as small as possible:

$$\mathbf{p}^* = \arg \min_{\mathbf{p}} \sum_{i=1}^n r_i(\mathbf{p})^2. \quad (2)$$

This is found using a Levenberg–Marquardt (LM) algorithm [7], [8], a standard iterative least-squares method that starts from the tag's previous position (or a guess when starting) and repeatedly refines the estimate until the total error stops decreasing. Each run needs to know how the residuals respond to the move of the estimate, captured by the Jacobian $\mathbf{J}$, whose i -th row is the unit vector pointing from anchor i toward the current estimate:

$$\mathbf{J}_i = \frac{\partial r_i}{\partial \mathbf{p}} = \frac{\mathbf{p} - \mathbf{a}_i}{\|\mathbf{p} - \mathbf{a}_i\|}. \quad (3)$$

Moving the estimate directly toward or away from an anchor changes that anchor's range the most, while moving sideways barely affects it. Using this sensitivity, each iteration computes a correction Δ to the estimate by solving

$$(\mathbf{J}^T \mathbf{J} + \lambda \mathbf{I}) \Delta = \mathbf{J}^T \mathbf{r}, \quad \mathbf{p} \leftarrow \mathbf{p} - \Delta. \quad (4)$$

The right side of the equation, $\mathbf{J}^T \mathbf{r}$, points in the direction that reduces the total error the fastest, and the matrix on the left scales that so it lands near the true minimum. The damping term λ controls the size of the step: the tag lowers λ to take large, coarse steps when the estimate is improving and raises it to take small, finer steps when it is not, which keeps the search smooth even with noisy data and conditions. Because the search iterates on the last estimated position, only a few steps are needed per update. Three noise-free ranges generically constrain the tag to two candidate points; a fourth anchor resolves the ambiguity and over-determines the system. If all anchors were on a flat single plane, it would leave the vertical coordinate z ambiguous (the rover could sit equally above or below the plane). Therefore, the anchors must each be deployed at different elevations such that a unique 3D solution exists.

III. METHODOLOGY

A. Hardware and Localization Pipeline

The testbed comprised four stationary UWB anchors and one roaming tag (Fig. 1), each a Makerfabs MaUWB_ESP32S3 module pairing a Qorvo DW3000 UWB transceiver with an STM32 ranging co-processor. The anchors and the tag used double-sided two-way ranging (TWR) exchanges at a 6.8 Mbps pulse rate on UWB channel 9 (8 GHz), and the tag solved its own three-dimensional position on-board from the resulting ranges using the warm-started Levenberg–Marquardt least-squares estimator of (2)–(4). Each position fix, together with the raw per-anchor ranges, was streamed over Wi-Fi to a logging server at a mean rate of 9.2 Hz.

The four anchors were fixed at different heights spanning roughly $1.56 \times 0.65 \times 1.42$ m (a convex-hull volume of 0.33 m^3), at the coordinates listed in Table I, so that the vertical coordinate was observable. For the tag position used

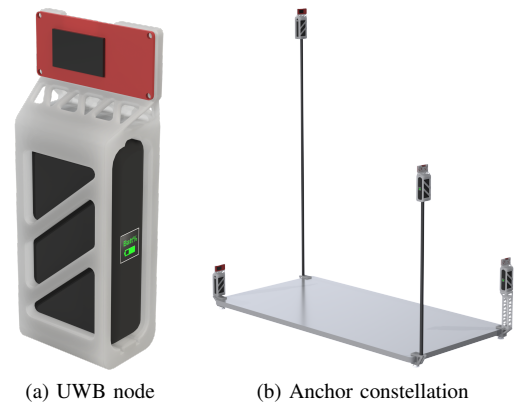


Fig. 1: Rendering of the localization testbed. (a) A single Makerfabs MaUWB_ESP32S3 node (Qorvo DW3000 + STM32) in its enclosure with an OLED telemetry display. (b) The four anchors A_0 – A_3 fixed at deliberately non-coplanar heights around the tag's workspace, so that the vertical coordinate is observable.

throughout, this constellation yields a geometric dilution of precision (GDOP) of 2.02, with a horizontal dilution of precision (HDOP) of 1.77 and a vertical dilution of precision (VDOP) of 0.98. Dilution of precision quantifies how the anchor geometry amplifies ranging error into position error, with lower values indicating a geometry that constrains position more tightly and a value near one indicating a near one-to-one transfer. GDOP is the combined three-dimensional factor, while HDOP and VDOP isolate its horizontal (x - y) and vertical (z) components. The cross-range (y) axis is the weakest of the three, reflecting the constellation’s 0.65 m y -span against 1.56 m and 1.42 m in x and z respectively.

TABLE I: Anchor Coordinates

Anchor	x (m)	y (m)	z (m)
A_0	0.00	0.00	0.00
A_1	1.56	0.00	0.77
A_2	1.56	0.65	0.20
A_3	0.00	0.65	1.42

All positions were measured to the antenna center with a tape measure, with an estimated survey uncertainty of ± 1 cm. The tag was held at a single measured ground-truth position (0.78, 0.32, 0.15) m for every trial, placing it 0.85–1.52 m from the individual anchors. Each anchor’s constant range offset was removed beforehand through a one-time bias calibration using the known anchor and tag positions. To prevent this calibration from masking obstruction effects, all trials were recorded in a *raw* mode, with position smoothing and outlier rejection disabled, so that a degraded link propagated directly into the reported position.

B. Trials

Each trial obstructed exactly one anchor–tag link by placing a test material in the path, roughly 5 cm from the tag, while the other three links remained clear. Five materials were evaluated, chosen to span the density, thickness, and reflectivity expected of lunar-relevant barriers. Each obstruction presented an in-path depth of approximately 10 cm and was positioned to block only the single link under test. Three of the materials shared a common plastic container so their effects could be compared directly: an empty container (a control for the act of placing an object in the path), the same container filled with dry leveling sand (a regolith analog), and the same container wrapped in aluminum foil. The foil case is therefore not a 10 cm slab of metal but two thin conducting layers — one on each face of the container — spanning that same depth, so it isolates the effect of a highly reflective surface from that of a thick absorptive medium. The remaining two materials were a paver stone and a solid concrete block. A clear-line baseline with no obstruction was also recorded. Every material was tested against each of the four anchors (A_0 – A_3) in turn, and each trial ran for 30 s, yielding roughly 277 position samples per trial.

C. Data Cleaning and Metrics

Twenty-five recordings were made in total. Before each recording, the obstruction’s placement was verified visually to confirm it intersected only the intended anchor–tag line of sight; trials in which this could not be confirmed were re-run before any data were examined. No trial was excluded on the basis of its recorded error. The logged ranges were also inspected to confirm that only the intended link was affected. For each material–anchor cell we retained the final take and discarded the superseded ones, leaving 21 trials (one baseline plus five materials $\times$ four anchors). All achieved a 100% fix rate (the signal was never lost). For each trial we computed the three-dimensional root-mean-square error (RMSE) of the position estimate against ground truth (the tape-measured true position), its horizontal (2D) and vertical (Z) components, and the median and 95th-percentile per-sample error. To isolate the ranging-level cause of any position error, we also computed, for the obstructed link, its mean *range bias* (the additional distance measured due to the obstacle). This range bias is taken against the surveyed ground truth, and is therefore distinct from the solver’s residual value $r_i(\mathbf{p})$ of (2), which is measured against the current position estimate as a whole rather than the true distance.

IV. RESULTS

A. Localization Error by Material

In the baseline, the system achieved a 3D RMSE of 5.1 cm (2D 4.8 cm, Z 2.0 cm), confirming centimeter-level accuracy in the unobstructed case. Obstructing a single link degraded accuracy as the density and thickness of the material class increased, as shown in Fig. 2 and Table II. The empty container raised 3D RMSE slightly, to 9.3 cm, confirming that the plastic container itself was mostly transparent to the UWB signal. Sand (17.3 cm), the paver stone (23.8 cm), and especially the concrete block (47.8 cm) produced progressively larger errors, the concrete block degrading accuracy by roughly a factor of nine relative to baseline. Despite being the most reflective material tested, the aluminum foil produced just 13.0 cm of error which was less than every solid or granular material, and only slightly above the empty container.

The choice of which anchor was blocked also mattered (Fig. 3). Error was largest when the obstruction fell on anchor A_2 for almost every solid material, reaching 74.7 cm for concrete. Neither the geometric amplification factor for A_2 (among the smallest in this constellation, Section IV-B) nor its distance from the tag (equal to that of A_0 within the survey uncertainty) accounts for this. The anchor-dependence is therefore not resolved by the present measurements. This dependence also expanded the variance within each material, visible as the error bars in Fig. 2.

B. Ranging Mechanism

The position errors are explained almost entirely by a single ranging-level quantity: the range measured to the blocked anchor. The anchor’s pulse was delayed and dispersed by the obstructions, so its estimated range increased; this injected

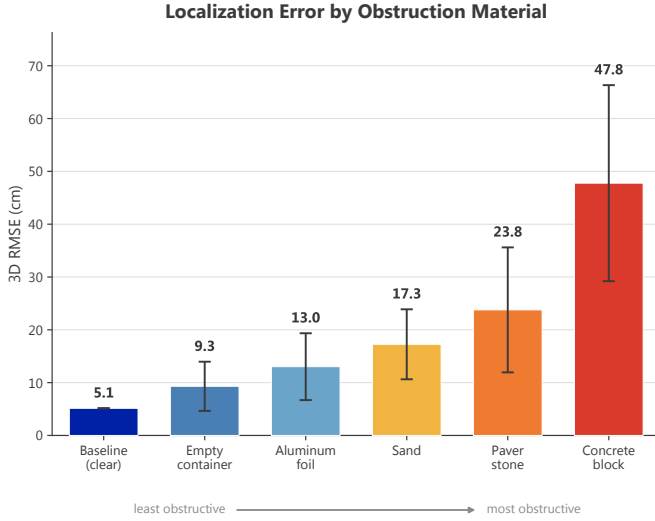


Fig. 2: Mean 3D localization RMSE when a single anchor link is obstructed, by material, averaged over the four blocked-anchor positions. Error bars show the standard deviation across anchors; the dashed line is the clear-line baseline. Error grows monotonically across the material classes tested.

TABLE II: Localization Error and Injected Range Bias by Material

Material	3D RMSE (cm)	2D (cm)	Z (cm)	P95 (cm)	Range bias (cm)	Range SD (cm)	Density (kg/m ³)
Baseline (clear)	5.1	4.8	2.0	9.7	—	—	—
Empty container	9.3	9.0	2.1	12.5	0.1	1.2	~243
Aluminum foil	13.0	11.5	5.9	16.3	7.6	2.6	~2000
Sand	17.3	15.5	6.6	20.7	15.2	1.4	~1500
Paver stone	23.8	21.2	7.6	27.5	16.7	1.6	~2000
Concrete block	47.8	43.1	19.4	52.9	42.5	2.9	~2100

Values are means over the four blocked-anchor positions. Range bias is the mean offset between the measured and true range on the obstructed link, and range SD is the standard deviation of the per-sample range error on that link. Density values are estimated for the material class, not for the specific specimens used; they indicate the density range and order rather than supporting quantitative comparison.

range bias grew with material class, from essentially zero for the empty container to 42.5 cm for concrete (Table II). Plotting each trial’s 3D RMSE against the range bias on its blocked anchor (Fig. 4) reveals a near-unity linear relationship,

$$\text{RMSE} \approx 5.8 \text{ cm} + 0.998 \Delta d, \quad (5)$$

with a correlation of $r = 0.92$, where Δd is the injected range bias. The near-unity slope is a property of this constellation rather than a general result. For a bias on anchor i , the least-squares position shift is $\|\mathbf{C}\mathbf{G}_i^T\|$ with $\mathbf{C} = (\mathbf{G}^T\mathbf{G})^{-1}$ and $\mathbf{G}$ the matrix whose rows are the unit line-of-sight vectors $\mathbf{J}_i$ of (3); evaluating this at the tag position gives amplification factors of 0.69, 1.36, 0.74, and 1.10 for A_0 through A_3 , a mean of 0.97. This mean is consistent with the observed slope of 0.998; the per-anchor factors, however, do not predict individual trials. Regressing the observed RMSE

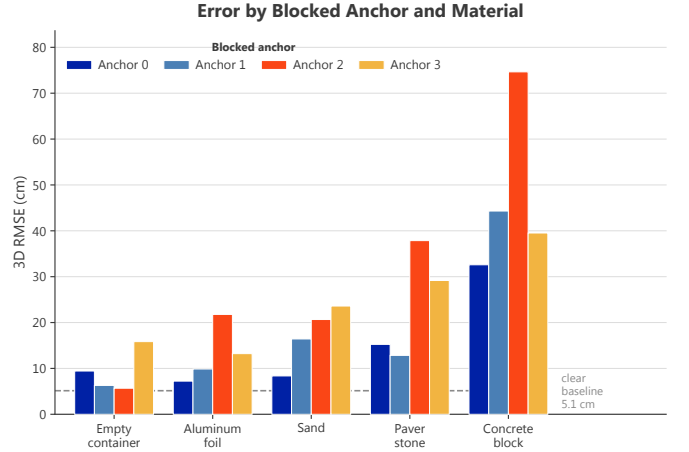


Fig. 3: 3D RMSE by material and by which anchor was blocked. Blocking A_2 produced the largest error for almost every solid material.

on (range bias) $\times$ (amplification factor) gives $r = 0.77$, below the $r = 0.92$ from range bias alone. Therefore, the aggregate agreement reflects the mean geometry rather than a per-link law. A more balanced constellation would distribute a single-link bias further across the remaining residuals and yield a smaller mean factor.

Equation (5) therefore implies that every additional centimeter of range bias on a single blocked link translates into approximately one centimeter of position error, across the 0–43 cm range of biases observed here, above the ~ 5.8 cm floor set by the system’s baseline noise. The obstruction’s effect on localization is mediated by the extra propagation delay it imposes, rather than by signal loss or dropped ranges. Every trial still reported a 100% fix rate. The empty container remains an outlier to (5): its mean range bias was 0.1 cm with only 1.2 cm of per-sample scatter, yet its 3D RMSE reached 9.3 cm. Neither quantity on the obstructed link accounts for the excess.

C. Error Distribution and Geometry

The obstructions degraded the horizontal and vertical components at proportionally comparable rates (Fig. 5): under concrete the Z RMSE rose to 19.4 cm from a 2.0 cm baseline (9.7 $\times$) while the 2D (XY) error rose to 43.1 cm from 4.8 cm (9.0 $\times$), the vertical error remaining smaller in absolute terms. The DOP decomposition of Section III identifies the cross-range (y) axis as the least well constrained, reflecting the constellation’s 0.65 m y -span; the vertical component, despite the tag sitting low at $z = 0.15$ m, was comparatively well conditioned (VDOP 0.98). The present design does not resolve a difference between the two. The full per-sample error distributions (Fig. 6) display how denser materials not only apply a constant offset but also raise the median error and increase the error ceiling, with the 95th-percentile error reaching 52.9 cm for concrete versus 9.7 cm at baseline. Concrete’s pooled distribution is visibly multimodal, with the

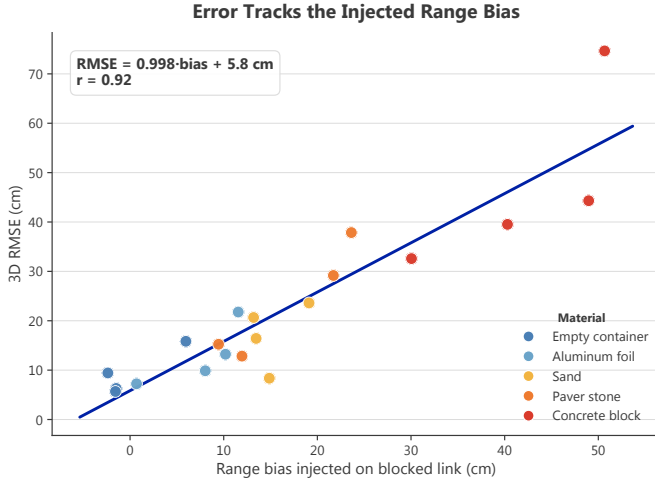


Fig. 4: Each trial’s 3D localization RMSE against the range bias injected on its obstructed link. The near-unity slope (0.998; $r = 0.92$) shows position error is driven almost one-to-one by the range bias an obstruction adds to a single link.

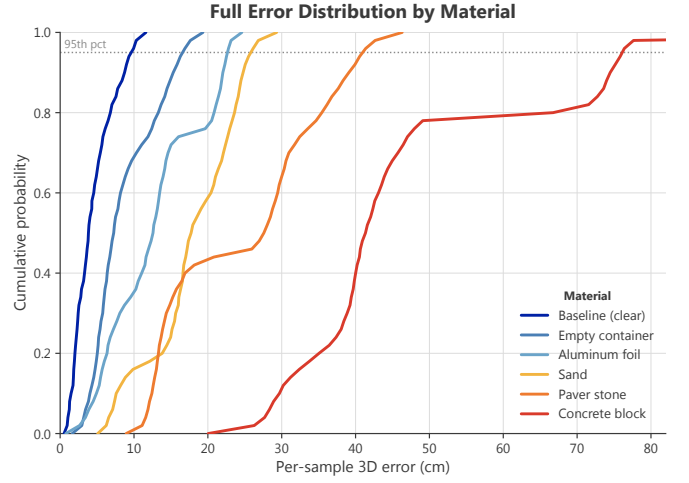


Fig. 6: Cumulative distribution of per-sample 3D error, pooled across the four blocked-anchor positions for each material. Denser materials shift the median and lengthen the upper tail; concrete’s pooled distribution is multimodal.

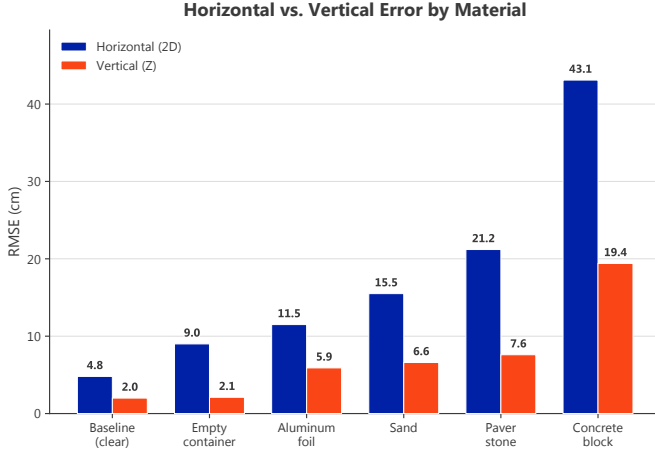


Fig. 5: Horizontal (2D) versus vertical (Z) RMSE by material, both components inflating at comparable proportional rates.

error concentrated in separated clusters rather than spread about a single mean.

V. DISCUSSION

The results confirm that physical obstruction of even a single anchor link meaningfully degrades UWB localization. An obstruction creates a source of excess propagation delay on a blocked link, injecting a positive range bias that the solver converts almost one-to-one into position error (5). Because three of the four links remained clear in every trial, the solver never lost a fix; instead it incorporated the range bias on the fourth link into the calculated position. This is the practically important failure mode for a lunar rover, which will routinely have some anchors shadowed by terrain or structures while others stay visible.

The results support the density prediction and reject the reflectivity prediction. Range bias increased monotonically across the material classes tested, from essentially zero for the two low-density cases to 42.5 cm for solid concrete (Table II). Because specimen densities were not measured directly, this ordering is qualitative: the trials establish that bulk material class dominates the injected delay, without resolving the functional form of that dependence. The density prediction was therefore tested against a material-class ordering used as a proxy, not against measured specimen densities. The concrete block, the densest material class tested, produced by far the largest bias and error, followed by the paver stone and sand. Reflectivity, however, was not the primary contributor. Aluminum foil, the most reflective material, produced less error than any of the granular or solid materials, because a thin conductor blocks the direct path but adds little excess path length. Two layers roughly $16\ \mu\text{m}$ thick spanning a 10 cm gap occupy on the order of 0.03% of the in-path volume, so the obstruction presents an effective bulk density close to that of the empty container despite its reflective surface. Because UWB’s multipath resistance filters out the reflections, the tag successfully recovers the direct-path estimate. The range-scatter column separates the two effects more cleanly than the mean bias alone. Foil injected the second-largest per-sample scatter (2.6 cm) while adding one of the smallest mean biases (7.6 cm), the reverse of the granular and stone obstructions, which added large mean biases with little scatter (sand and paver both ≤ 1.6 cm). This split is consistent with reflectivity perturbing the *variance* of the first-path estimate while bulk density shifts its *mean*; concrete, the densest and most disruptive material, raised both. The result that sand — the closest analog to lunar regolith — inflated error to 17.3 cm suggests that a rover partially buried in or shadowed by regolith would suffer decimeter-scale errors, which is

significant for autonomous construction tolerances.

There are many direct implications from these findings. First, because the damage is concentrated in the injected range bias on individual links, obstruction detection by flagging and de-weighting a range whose residual departs from the solver consensus is a promising mitigation. The raw ranges logged here already expose that signal. Second, dilution of precision predicts that a single-link bias is absorbed most poorly along the constellation's least-constrained axis, so anchor placements for uneven lunar terrain should avoid a short baseline in any one direction. The present trials motivate but do not test this directly.

A. Limitations

This is a controlled benchtop study and its scope should be read accordingly. The tag was stationary and measured at a single position, so the results demonstrate static ranging accuracy rather than dynamic tracking. The test volume was small, and the links were sub-2 m. A full lunar network would be much larger scale. Each material–anchor cell was only measured once, so each material's four measurements carry limited statistical power and give no repeatability estimate. The trials were conducted at room temperature in air, omitting the vacuum, temperature extremes, and electrostatically charged regolith of the lunar surface.

Each anchor's range offset was calibrated at the same single position at which the system was subsequently evaluated, so the 5.1 cm baseline represents a best case that would not necessarily hold across the workspace.

The obstruction was placed 5 cm from the tag, close enough that antenna coupling cannot be fully excluded, whereas the lunar scenario motivating this work involves terrain shadowing an anchor far from both endpoints; the effect of obstruction position along the link was not varied.

Received signal power was not logged, so the conclusion that delay rather than attenuation dominates is inferred from the 100% fix rate and the sign of the range bias rather than measured directly. In addition, the ~ 277 samples per trial are strongly autocorrelated — the tag was stationary and the solver warm-started from its previous estimate — so the effective sample size, and hence the precision of each per-trial RMSE, is substantially smaller than the raw count suggests.

Specimen densities were not measured; materials were selected to span a qualitative range of bulk density and the reported ordering is by material class.

Finally, only single-link obstructions were studied. Simultaneous obstruction of multiple links, which would threaten the tag's ability to retain a fix rather than merely degrade its accuracy, remains untested.

VI. CONCLUSION

This study quantified how physical line-of-sight obstructions degrade UWB two-way ranging localization under conditions relevant to lunar rovers. In a four-anchor, single-tag testbed, obstructing a single anchor link raised three-dimensional RMSE from a 5.1 cm clear-line baseline to between 9.3 cm

for an empty container and 47.8 cm for a solid concrete block, with denser and thicker material classes producing consistently larger errors. The degradation is dictated almost entirely by the propagation delay an obstruction imposes. It injects a positive range bias on the blocked anchor, which the trilateration solver converts into position error at a near-unity slope of 0.998 ($r = 0.92$), above a ~ 5.8 cm noise floor. Reflectivity was not the primary driver. The thin aluminum foil produced less error than any granular or solid material. Therefore, material class and thickness, not reflectivity alone, drive the error. Because fixes were never lost while three links stayed clear, the important failure mode is incorporating a positive bias on a degraded link rather than an outright dropout. These results motivate two ways to mitigate these issues in a deployed lunar network: per-link obstruction detection that de-weights a link whose residual departs from consensus, and anchor constellations that avoid a short baseline in any one direction, since the dilution of precision predicts that a single-link bias is absorbed most poorly along the least-constrained axis — a design principle this study motivates but, at $n = 1$ per configuration, does not itself validate. Extending the study to dynamic tracking, logging of channel impulse response and first-path power to separate direct-path attenuation from first-path detection failure, larger link distances, simultaneous multi-link obstruction, and the vacuum and thermal extremes of the lunar environment remains future work.

DATA AVAILABILITY

The raw per-anchor range logs and position fixes for all 21 trials are available from the author on request.

The code used in the server software, as well as the firmware for the micro-controllers are also available upon request.

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During the preparation of this work, the author used Claude (Anthropic) to assist with $\LaTeX$ and BibTeX formatting, with language editing of the localization model described in Section II, and with statistical analysis of the trial data, generation of the figures, and drafting of the Results and Discussion sections. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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